

MAYUR NARWADE

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PROFESSIONAL SUMMARY

AI/ML Engineer and Computer Science graduate with hands-on experience in healthcare AI, clinical NLP, computer vision, deep learning, REST API development, and Generative AI. Contributed to a Medication Error Detection System using OCR, ClinicalBERT, scispaCy, RxNorm-based normalization, rule-based validation, XGBoost, FastAPI, and healthcare databases. Experienced with Python, TensorFlow, PyTorch, OpenCV, Hugging Face Transformers, RAG, Docker, and Google Cloud Vertex AI.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, SQL, HTML, CSS | **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, CNN, TensorFlow, PyTorch, Hugging Face Transformers, RAG

Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, MongoDB, MySQL | **Tools & Cloud:** Git, Docker, Streamlit, Google Cloud Platform, Vertex AI, VS Code

EXPERIENCE

Technical Intern — National Institute of Electronics & Information Technology (NIELIT), Chhatrapati Sambhajinagar | Feb 2026 – Present

- Contributed to healthcare AI development, backend integration, technical research, testing, documentation, and workflow improvement in a government R&D environment.
- Supported FastAPI services, API implementation, debugging, database workflows, dataset preparation, and validation while maintaining data confidentiality and institutional standards.
- Mentored 90+ students during Advanced IoT and Microcontrollers bootcamps and supported technical and institutional activities.

PROJECTS

Medication Error Detection System (MEDS) | Python, FastAPI, ClinicalBERT, scispaCy, OCR, XGBoost, RxNorm, PostgreSQL, MongoDB

- Contributed to an AI-assisted healthcare system that identifies potential medication errors by analyzing prescriptions, patient information, medicine knowledge, and clinical rules.
- Supported a hybrid pipeline combining OCR, clinical NLP, RxNorm-based normalization, rule-based validation, and XGBoost inference for interaction, allergy, dosage, and duplicate-medication risk analysis.
- Assisted with REST API development, database integration, testing, debugging, role-based access workflows, JWT authentication, and technical documentation.

Driver Drowsiness Detection System (DDS) | Python, OpenCV, Computer Vision, Facial Landmarks, NVIDIA Jetson Nano

- Developed and tested a real-time computer-vision pipeline that monitors eye behavior and prolonged eye closure to detect potential driver drowsiness.
- Applied temporal thresholds to distinguish normal blinking from sustained fatigue and integrated an automated alert mechanism.
- Optimized edge-device testing to achieve facial-landmark processing below approximately 45 ms while sustaining up to 30 FPS.

Flowerpedia — AI-Powered Flower Classification Application | Python, TensorFlow, CNN, ResNet50, Transfer Learning, Streamlit

- Built a deep-learning application for classifying 102+ flower species using the Oxford 102 Flowers dataset and ResNet50 transfer learning.
- Implemented image preprocessing, augmentation, feature extraction, confidence scoring, and an interactive Streamlit interface for real-time predictions.
- Achieved over 95% classification accuracy during project evaluation.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — MGM College of Engineering & Technology, Nanded | Nov 2022 – Jul 2026

CGPA: 7.5/10.0

CERTIFICATIONS

Google Cloud: Inspect Rich Documents with Gemini | Multimodal RAG | Prompt Design in Vertex AI | Machine Learning I — Core Concepts

PUBLICATION

Zero-Shot Learning Based Multilingual Medical Symptom Understanding and Disease Inference System Using NLP and Semantic AI

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